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Research and Application of Quantum Computing Algorithm for Asset Portfolio Optimization

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***Abstract*—Portfolio optimization is a critical problem in the field of finance, and quantum computing offers a new approach to address this problem. The author explains the development process of the quantum computing model, including quantum bits, quantum entanglement, quantum gates, quantum circuits, while specifically focusing on quantum state preparation, problem coding, quantum gate operation and quantum measurement in the design of quantum algorithms. Furthermore, the author also introduces the mathematical model of portfolio optimization, including expected return, risk analysis, and mathematical constraints, and illustrates the performance evaluation results of quantum algorithms in portfolio optimization from the perspective of speed and optimization accuracy. With respect to traditional methods, quantum algorithms offer both speed and optimization accuracy advantages, and the optimization accuracy reaches 100%, which illustrates that the quantum algorithm has shown good performance in asset portfolio optimization.**

Keywords—Quantum computing, asset portfolio optimization, risk management, market adaptation

I. Introduction

In the financial field, portfolio optimization is a key strategy to achieve a balance between risk and return. With the rise of quantum computing, a new calculation method has appeared in this field, which provides a new way for portfolio optimization.

In the existing literature, although there have been studies on the potential application of quantum computing in the financial field, most of these studies focus on theoretical discussion and preliminary exploration. However, these studies have not yet formed a mature system in the algorithm and strategy of quantum computing applied to financial asset portfolio optimization. In this paper, a new algorithm framework combining quantum computing and portfolio optimization is proposed. By constructing quantum computing model and designing corresponding quantum algorithm, quantum computing technology is applied to portfolio optimization, which provides a new idea for financial innovation.

This paper first introduces the research background, motivation and objectives, and then discusses the application status and challenges of quantum computing in the financial field. Then, this paper introduces the construction of quantum computing model, the design of quantum algorithm and the application of quantum algorithm in asset portfolio optimization. Finally, this paper summarizes the performance evaluation results of quantum algorithm in portfolio optimization, and puts forward the prospect of future research direction.

II. Related work

In the financial field, asset portfolio optimization is a key strategy to achieve a balance between risk and return. Based on the fact that the improvement of education has changed the economic decision-making of China's residents and has also had a significant impact on the diversity of China's household financial asset portfolio, Hong Congkai used the data from the 2018 China Resident Income Survey Program to explore the impact of the improvement of education on the diversity of household financial asset portfolios [1]. Zhou Chunyang constructed a portfolio that not only included traditional stock assets and risk-free assets, but also introduced variance/volatility assets, and gave an explicit solution for the dynamic optimal portfolio, and finally found that the price of variance/volatility assets was an important state variable affecting the weight of the portfolio [2]. Qian Fucheng constructed a model of investor portfolio end-period wealth expected utility, in the case of two types of investor portfolio end-period wealth expected utility was equal, the value of market information was measured by the difference in the initial wealth input of the portfolio, and analyzed the relationship between the value of market information and the market volatility, the investment period, and the degree of investor risk aversion [3]. Fan Guobing analyzed and described the mathematical characteristics of the return, risk, and correlation risk of the risky asset portfolio, and introduced the asset allocation analysis model of the optimal risky asset portfolio, and used the data analysis software to carry out the auxiliary analysis of the mathematical characteristics, to analyze the theory of the risky asset portfolio by the key numerical analysis of the characteristics of the risky asset portfolio and to carry out the empirical analysis [4]. Zeng Yihang started from the modern asset portfolio theory, introduced the birth and causes of diversification theory, and at the same time introduced the centralized investment strategy, through the analysis of typical investment cases, discussed the risks and returns of investment strategies, and gave investors suggestions on the pros and cons of choosing [5].

Not only that, Ollitrault P J discussed the recently introduced quantum algorithms for wave packet evolution in the first quantization and exploited the potential quantum advantage of mapping its spatial lattice representation to logarithmically many quantum bits, and also reviewed the scalar properties of two alternative time evolution algorithms, namely, the variational quantum algorithm (based on the McLachlan variational principle) and the conventional

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Trott-type evolutionary algorithm (based on the Lee-Trotter-Suzuki formula) [6]. An D solved the quantum linear system problem by adiabatic quantum computation (AQC) as a method through an optimized scheduling function, and numerical results showed that the quantum approximation optimization algorithm yielded the lowest runtime when compared to time-optimized AQC, vanilla AQC, and recently proposed randomized methods [7]. Suaau A demonstrated experimentally that certain partial differential equations can still be solved on a quantum computer even if an error-correcting quantum chip was not available in the short term [8]. Ajagekar A introduced the basic concepts of quantum computing while pointing out the limitations of current quantum computers and also provided an overview of quantum algorithms, setting expectations for future quantum devices by exploring similar applications that may benefit from quantum algorithms implemented on such devices [9]. Graham T M demonstrated several quantum algorithms on a programmable gate model neutral atom quantum computer using a tightly focused beam of light scanning a two-dimensional array of quantum bits, with the final results highlighting the emerging capabilities of neutral atom quantum bit arrays for general-purpose programmable quantum computation as well as the preparation of nonclassical states for quantum-enhanced sensing [10]. Gills had comprehensively sorted out and classified the literature of quantum computing through systematic review [11]. Blunt N S discussed its application in drug discovery from the perspective of quantum computing [12]. Herman D discussed the application of quantum computing in the financial field, including risk assessment, portfolio optimization, market analysis and so on [13]. Ghosh M proposed a new quantum ant colony optimization algorithm, described the design and implementation of quantum algorithm, and verified its effectiveness in solving optimization problems through experiments [14]. Weigold M studied the coding mode in quantum algorithm and analyzed the influence of different coding strategies on the performance of quantum algorithm [15]. Although the above studies provide insights into the diversity of financial asset portfolios and the application of quantum computing in finance for this research, a sophisticated set of algorithms and strategies have not yet been explored to address the specific challenges of financial asset portfolio optimization. This study explores the application of quantum computing algorithms in financial asset portfolio optimization, evaluates their performance, analyzes their performance under different market conditions, and develops quantum optimization strategies that are more adapted to the characteristics of financial markets.

III. Methods

A. Quantum Computing Model

Quantum computing model is a cutting-edge technology in the research of asset portfolio optimization, which is based on the principle of quantum mechanics and provides a new way of data processing. In this model, quantum bits (qubits) are the basis for building quantum algorithms. Unlike classical bits, qubits are able to exist in a superposition of states of 0 and 1 at the same time, a property that greatly increases the parallelism of computation. A single qubit can be represented as a vector in a two-dimensional Hilbert space with a superposition of states:

$$|\psi\rangle=\alpha|0\rangle+\beta|1\rangle \tag{1}$$

$|\psi\rangle$ refers to the superposition state of a quantum bit, and α and β refer to the complex amplitudes of the ground states $|0\rangle$ and $|1\rangle$, respectively, in the superposition state.

Quantum entanglement describes a special kind of correlation between two or more qubits such that even if they are spatially far apart, a change in the state of one qubit instantly affects the other entangled qubits. This non-deterministic property allows quantum computers to consider complex interactions between multiple assets simultaneously in asset portfolio optimization. In this case, the entangled states of two qubits can be expressed as:

$$|\Psi\rangle=\frac{1}{\sqrt{2}}(|00\rangle+|11\rangle) \tag{2}$$

$|\Psi\rangle$ is the entangled state of two quantum bits, and $|00\rangle$ and $|11\rangle$ are the superposition states of two quantum bits, respectively.

Quantum gates are the basic logic units that operate qubits, they change the state of the qubits by means of the Missy transformations. Quantum gates include Pauli-X gates, Pauli-Y gates, Pauli-Z gates, Hadamard gates, CNOT gates, etc., and each of these gates has its own specific function, e.g., Hadamard gates are used for the creation of superposition states, and CNOT gates are used to create entanglement. The matrix form of Pauli-X gate, Hadamard gate, CNOT gate are respectively:

$$X=\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix} \tag{3}$$

$$H=\frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix} \tag{4}$$

$$CNOT=\begin{pmatrix}1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1\\ 0 & 0 & 1 & 0\end{pmatrix} \tag{5}$$

X refers to the Pauli-X gate, H refers to the Hadamard gate which creates an equal probability superposition state, and $CNOT$ refers to the CNOT gate used to create entangled states.

The quantum line model forms a quantum line by connecting quantum gates in a specific order to perform complex computational tasks. It is represented by a series of quantum gates connected in series, each corresponding to a youngest-positive transformation, and the entire line can be represented as a composite transformation U which is the product of the youngest-positive matrices of all the gates:

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$$U=U_n U_{n-1} \cdots U_2 U_1 \tag{6}$$

U_n and U_{n-1} are the youngest transformations of the n and $n-1$ quantum gates, which are multiplied in the order in which they are in the line.

The model allows quantum algorithm designers to construct and simulate quantum algorithms in an intuitive way that allows optimization of asset portfolios on a quantum computer.

B. Quantum Algorithm Design

The quantum state preparation uses a Hadamard gate to operate on each quantum bit to transition it from a deterministic $|0\rangle$ state to a superposition state where the two complex coefficients α and β represent the probability amplitudes of the quantum bit being in $|0\rangle$ and $|1\rangle$, respectively, and satisfy the normalization condition $|\alpha|^2+|\beta|^2=1$.

Next, quantum gate operation is used to simulate different parameters and constraints in portfolio optimization. For example, Pauli-X can represent whether to buy or sell assets, and CNOT gate can simulate the correlation between assets. In this process, the sequence and parameters of quantum gate must be carefully selected to ensure that quantum gate can accurately represent the return, risk and correlation of portfolio. Quantum entanglement involves information sharing between qubits, resulting in entangled states. The next part of the algorithm involves problem coding, which codes a specific portfolio optimization problem in a way that qubits can handle. Specifically, the characteristics of each asset, such as expected return and volatility, are encoded into the phase and amplitude of the quantum state. Once the algorithm is executed, the quantum state is measured, which will cause the quantum state to collapse to a specific ground state. Therefore, quantum states represent a specific portfolio allocation. Once the quantum state is obtained, it must be measured by a classical computer to evaluate its effectiveness as a portfolio allocation.

C. Mathematical Model for Asset Portfolio Optimization

In constructing a mathematical model for asset portfolio optimization, the first step is to define the expected return, which is obtained by calculating the sum of the products of the expected returns of all the assets in the portfolio and their respective weights. Specifically, a portfolio containing n assets, with asset i having a weight of w_i and its expected return of $E[R_i]$, has an expected return $E[R_p]$ of the entire portfolio:

$$E[R_p]=\sum_{i=1}^{n} w_i \cdot E[R_i] \tag{7}$$

Risk assessment is based on the standard deviation of portfolio returns, reflecting the volatility of asset returns. The risk of a portfolio of assets can be expressed as the square root of the trace of the covariance matrix of the returns of all the assets in the portfolio, i.e.:

$$\sigma_p=\sqrt{\mathrm{Tr}(\Sigma \cdot W \cdot W^T)} \tag{8}$$

Where σ_p is the standard deviation of the asset portfolio returns used as a measure of risk, Σ is the asset return co-matrix, W is the vector of asset weights, Tr is the trace of the matrix, i.e. the sum of the diagonal elements of the matrix, and W^T is the transpose of W.

Constraints include budget constraints, i.e., the sum of all asset weights must be equal to one, and upper and lower bounds on individual asset weights:

$$\sum_{i=1}^{n} w_i=1 \tag{9}$$

$$w_{i\min} \le w_i \le w_{i\max} \tag{10}$$

w_i is the weight of asset i in the portfolio, $w_{i\min}$ is the minimum weight of asset i, $w_{i\max}$ is the maximum weight of asset i, and n is the total number of assets. These constraints ensure that the portfolio's weight allocations are consistent with the investor's overall capital allocation strategy while also meeting exposure limits on individual assets.

D. Quantum Algorithm Implementation

In the quantum computing algorithm implementation for asset portfolio optimization, the quantum programming language Qiskit is used to materialize the quantum circuit design. A quantum circuit object containing the corresponding number of quantum bits is initialized in Qiskit to represent the state of each asset. Each quantum bit is converted to a superposition state via a Hadamard gate, and subsequently, a parameterized revolving gate is applied via .rx() for the expected return and risk characteristics of each asset. To model the correlation between assets, a CNOT gate is used to create quantum entanglement. Entanglement is created between assets via .cx(0, 1).

During the implementation of a quantum algorithm, quantum control logic is integrated to ensure that specific constraints are met. While specific quantum algorithm implementation details can be complex, Qiskit provides powerful tools and libraries to help build and test quantum circuits. After completing the quantum circuit design, Qiskit's Aer simulator is used to simulate the execution of the quantum algorithm. The quantum circuit is submitted to the simulator through the execute() function and a certain number of shots is set to obtain statistically reliable results. The results are shown in Table I:

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TABLE I. CIRCUIT DESIGN SIMULATION RESULTS

Experiment ID	Qubits	Initial Assets	Hadamard Gates	Parameterized Rotation Gates	CNOT Gates	Simulator Shots	Measurement Results Example	Feasible Configurations Ratio
1	5	10	5	10	4	1024	[01011, 11001]	75%
2	10	20	10	20	9	2048	[11010, 01010]	80%
3	15	30	15	30	14	4096	[11100, 00011]	85%
4	20	40	20	40	19	8192	[10101, 01111]	90%
5	25	50	25	50	24	16384	[11110, 10101]	92%

Table I visualizes the simulation performance of quantum computing algorithms in asset portfolio optimization through key metrics. These metrics not only reveal the computational efficiency and optimization capability of the algorithm, but also provide data support for further algorithm improvement and financial strategy formulation.

IV. RESULTS AND DISCUSSION

A. *Algorithm Performance Evaluation Results*

The quantum algorithm realizes the simultaneous evaluation of a large number of possible asset configurations through quantum superposition and quantum entanglement, and in the process of researching and applying quantum computing algorithms for asset portfolio optimization, this paper evaluates the speed and optimization accuracy performance of the quantum algorithms and the traditional methods, and the evaluation results are shown in Figures 1 and 2, respectively:

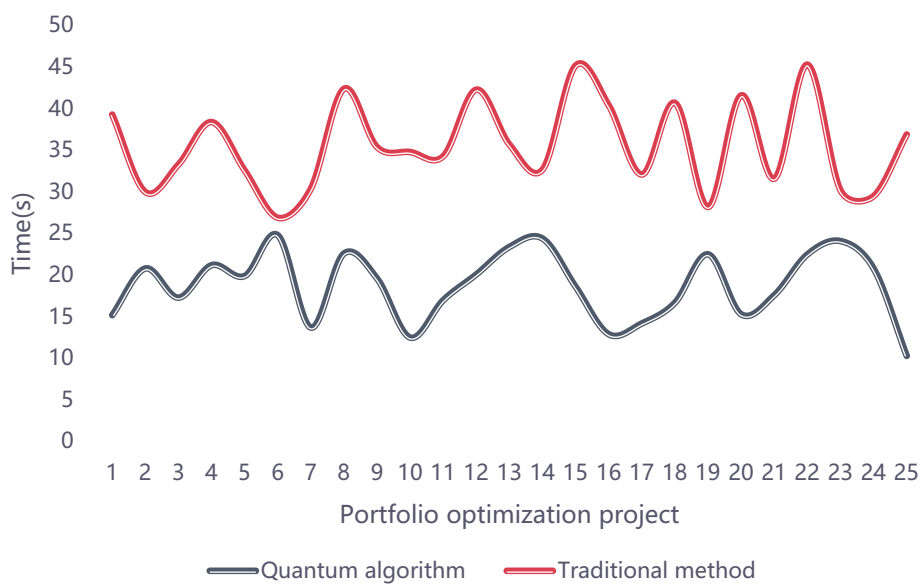

Figure 1. Optimizing speed

According to Figure 1, the asset portfolio optimization time of quantum algorithms is always lower compared to traditional methods, reaching as low as 10.3 s. This result indicates that quantum algorithms can significantly reduce the required computation time when dealing with high dimensional and large-scale optimization problems. The fast computational power enables quantum algorithms to respond to market changes in a timely manner and provide investors with more rapid and accurate asset allocation recommendations.

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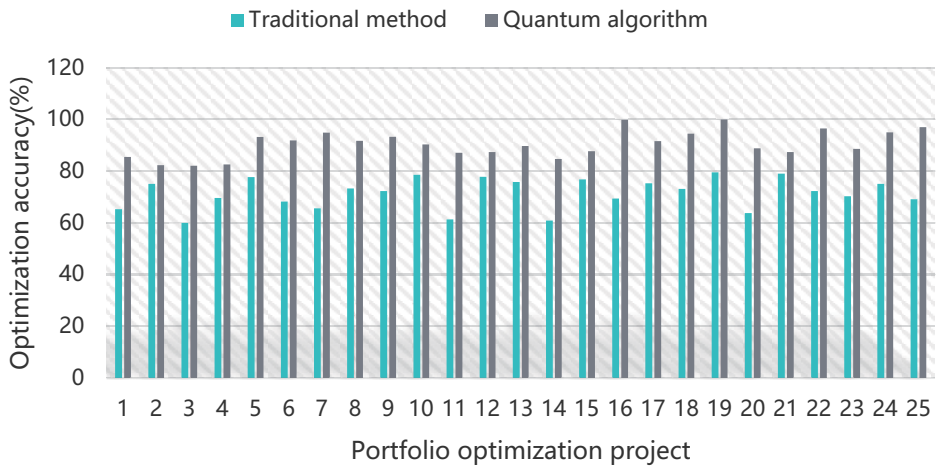

Figure 2. Optimizing accuracy

It is clearly pointed out in Figure 2 that the optimization accuracy of the quantum algorithm reaches up to 100%, which is much higher than that of the traditional methods, and this high accuracy can be attributed to the unique search mechanism of the quantum algorithm. Quantum algorithms utilize quantum superposition to explore many regions of the solution space simultaneously in polynomial-time and entanglement increases the algorithm's efficiency to search in high-dimensional space.

B. *Analysis of Market Condition Adaptability*

Quantum algorithms have shown impressive resilience and adaptability to market changes in portfolio optimization. The quantum superposition state allows processing multiple portfolios concurrently, which allows for quicker responses to market changes. Quantum algorithms also exhibit resilience in the investment portfolio. Whether the expected end result is short-term market arbitrage opportunity or long-term growth strategy, quantum computing can develop tailor-made solutions. By adjusting quantum wires and optimizing objectives, quantum algorithm can also divide assets according to different investment time ranges, ensuring that the performance of portfolio remains the best in the whole market cycle. In the current research, the adaptability of quantum algorithm is measured according to Sharp ratio, and the result is shown in Figure 3:

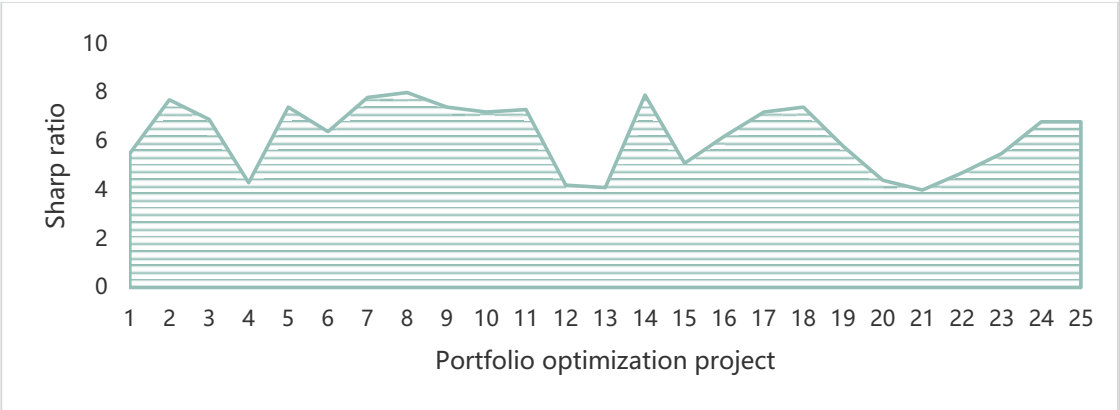

Figure 3. Sharpe ratio

Sharp ratio is a measure of how many times a strategy is superior to risk-free investment. The Sharp ratio of the quantum algorithm in Figure 3 in 25 portfolio optimization experiments is high, reaching a maximum of 8. In addition, quantum algorithm performs well in the field of financial risk management. Quantum algorithm effectively performs risk analysis, including portfolio value at risk (VaR) and conditional value at risk (CVaR). Investors can more accurately understand the potential losses of financial portfolios, especially under extreme conditions. Although quantum algorithms are promising, there are still some limitations due to quantum hardware. The stability, error rate and scalability of quantum bits are the main challenges in the research of quantum computers.

C. *Discussion*

This paper explores the application of quantum computing algorithm in portfolio optimization, and shows a new framework that combines quantum computing with financial strategy. The superior performance of quantum algorithm is attributed to its inherent ability, quantum superposition and entanglement. Quantum superposition can evaluate multiple potential asset allocation at the same time, and entanglement enhances the ability of the algorithm to

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explore high-dimensional solution space efficiently. These properties make quantum algorithm superior to traditional methods, especially when dealing with large-scale and high-dimensional optimization problems.

The ability to quickly and accurately optimize asset portfolios leads to more efficient capital allocation, lower risk exposure and potentially higher returns. This study also has limitations. The current state of quantum hardware is characterized by stability, error rate and scalability, which poses a great challenge to the practical application of quantum algorithm in finance. In addition, the complexity of quantum algorithm requires high-level professional knowledge, which combines deep financial knowledge with mastery of quantum programming and mechanics, which raises the threshold for financial professionals to use quantum computing in strategy.

In order to meet these challenges, future research will focus on enhancing the stability and error rate of quantum hardware, thus improving the overall performance and reliability of quantum computing systems. Taking educational measures to train financial professionals in quantum computing and cultivate a workforce capable of developing and implementing quantum algorithms in financial strategies. In addition, efforts should be made to combine quantum algorithm with traditional financial model, and make use of the advantages of the two methods to achieve more effective portfolio optimization.

V. Conclusion

This paper explores the application of quantum computing algorithms in the field of asset portfolio optimization, and proposes a novel algorithmic framework combining quantum computing and asset portfolio optimization. By constructing a quantum computing model and designing the corresponding quantum algorithm, this paper successfully applies quantum computing technology to the asset portfolio optimization problem, providing an innovative solution. The experimental results show that compared with the traditional methods, the quantum algorithm has significant advantages in optimization speed and accuracy, and can quickly respond to market changes and achieve real-time optimization of asset portfolios.

This paper addresses the computational efficiency and accuracy issues in asset portfolio optimization, especially when dealing with large-scale and high-dimensional datasets, quantum algorithms show the potential to outperform traditional computational methods. However, this paper also has some limitations. First, the stability, error rate, and scalability of quantum hardware are still the main challenges currently faced, which limits the application of quantum algorithms in practical financial fields. Secondly, the complexity of the realization of quantum algorithms requires experts in the financial industry not only to have profound financial knowledge, but also to master the principles of quantum programming and quantum mechanics, which undoubtedly increases the technical threshold. In addition, the interpretability of quantum algorithms should not be ignored, and the transparency and interpretability of algorithms are strictly required in the financial decision-making process.

Advances in quantum hardware technology and further optimization of quantum algorithms are expected to gradually overcome the existing challenges and realize the widespread application of quantum computing technology in the financial industry. Future research should strive to strengthen the research on the stability and error rate of quantum algorithms to improve the performance and reliability of quantum computers; promote quantum computing education and training to cultivate more financial industry professionals who master quantum computing technology; and explore the integration of quantum algorithms with traditional financial models to realize complementary advantages.

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